AI Project Voice Controlled Virtual Assistant for Visually Impaired Users

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Objective

The objective of this project is to develop an AI-based voice-controlled virtual assistant tailored for visually impaired users. The system will allow users to interact with digital devices through voice commands to perform tasks such as sending messages, making calls, browsing the internet, and accessing real-time information like weather and time. The system focuses on accessibility, ease of use, and natural language understanding.

Modules

1. **Voice Command Interface**
   This module captures voice input from the user using a microphone. It filters ambient noise and uses speech-to-text conversion to process commands in real-time.
2. **Natural Language Processing (NLP)**
   After converting speech to text, this module interprets user intent using NLP libraries and models. It breaks down commands into actions and parameters for appropriate responses.
3. **Task Execution Module**
   This module handles the execution of commands such as reading text messages, announcing the current time, searching online content, or navigating apps. It supports integration with device features and APIs.
4. **Text-to-Speech (TTS) Response**
   The system responds to users through spoken messages. Using a TTS engine, it converts textual responses back into natural-sounding speech for auditory feedback.
5. **User Profile and Personalization**
   Each user has a profile where preferences are stored (e.g., preferred voice, language, common tasks). The system learns from past interactions to offer personalized suggestions and faster response times.
6. **Offline Mode and Data Handling**
   This module ensures that certain basic functions work even without an internet connection. The system handles offline queries like time, reminders, and calendar events while syncing with the cloud when online.
7. **Security and Privacy**
   The assistant ensures privacy by locally processing sensitive voice data whenever possible. It offers options to enable or disable listening and includes voice authentication for security.

Tools and Technologies

• Programming Languages: Python, JavaScript
• Speech Recognition: Google Speech API, CMU Sphinx
• NLP: spaCy, NLTK, Transformers (Hugging Face)
• TTS: gTTS, pyttsx3

• Backend: Flask or Node.js
• Frontend (for admin settings): React or Electron.js
• Device Integration: Android SDK, Twilio (for calls/SMS)
